

Chapter 3

Complex Contour Integration

Definition 3.1. The *primitive* of a function $f : \mathcal{D} \rightarrow \mathbb{C}$ is an analytic function $F : \mathcal{D} \rightarrow \mathbb{C}$ such that $F' = f$ for any $z \in \mathcal{D}$.

Motivation 3.2. In the case of $\mathcal{D} \subseteq \mathbb{R}$, the first fundamental theorem of calculus (Theorem B.127) guarantees the existence of the primitive of any function $f : \mathcal{D} \rightarrow \mathbb{R}$, since we can construct a suitable integral as the primitive of f . However, this is not true in the case of $\mathcal{D}$ being an open subset of $\mathbb{C}$: a function that has a primitive must be analytic, c.f. Theorem 3.83. Therefore, it is natural to ask the question whether the primitive of a function $f : \mathcal{D} \rightarrow \mathbb{C}$ exists; this question is a key problem in our extension of $\mathbb{R}$ to $\mathbb{C}$.

Roadmap 3.3. This chapter mainly concerns the equivalence of the following four statements.

- (a) f has a primitive.
- (b) The complex integral of f on a closed curve vanishes.
- (c) The complex line integral of f is path independent.
- (d) f is analytic.
- (e) f is representable as a power series in each open disk fully contained in its domain.

After examining a stepping-stone integral in Definition 3.4, we formally define the concept of complex line integrals in Section 3.2 and prove Theorem 3.16 that states (a) $\Rightarrow$ (b). One key result of Section 3.3 is Theorem 3.40, which states that (a) $\Leftrightarrow$ (b) $\Leftrightarrow$ (c). The central result of Section 3.3 is the Cauchy theorem, which states (d) $\Rightarrow$ (b) in Theorems 3.45, 3.53, 3.57, 3.62, and 3.71 on various domains. In Section 3.4, the generalized Cauchy integral formula (Theorem 3.83) gives (a) $\Rightarrow$ (d); this is utilized in the proof of the Morera theorem 3.91, which states (b) $\Rightarrow$ (d). The equivalence of (e) to (d) is proved in Section 3.5.

3.1 A stepping stone

Definition 3.4. A function $f : [a, b] \rightarrow \mathbb{C}$ with $a, b \in \mathbb{R}$ is called *integrable* iff $\operatorname{Re} f, \operatorname{Im} f : [a, b] \rightarrow \mathbb{R}$ are Riemann integrable as in Definition B.106; then

$$\int_a^b f(x) dx = \int_a^b \operatorname{Re} f(x) dx + \mathbf{i} \int_a^b \operatorname{Im} f(x) dx. \quad (3.1)$$

Corollary 3.5. Denote by $\mathcal{V}$ the set of continuous functions $[a, b] \rightarrow \mathbb{C}$. Then $\mathcal{V}$ is a vector space and the integral $\int_a^b$ defined in (3.1) is a linear functional on $\mathcal{V}$.

Proof. We need to show that the integration is a linear map from $\mathcal{V}$ to $\mathbb{C}$, which holds because

$$\begin{aligned} \int_a^b (f(x) + g(x)) dx &= \int_a^b f(x) dx + \int_a^b g(x) dx; \\ \forall \lambda \in \mathbb{C}, \int_a^b \lambda f(x) dx &= \lambda \int_a^b f(x) dx. \end{aligned} \quad \square$$

Lemma 3.6. If $f : [a, b] \rightarrow \mathbb{C}$ is continuous and F is a primitive of f , i.e. $F' = f$, then

$$\int_a^b f(x) dx = F(b) - F(a). \quad (3.2)$$

Proof. This follows from (3.1) and the second fundamental theorem of calculus (Theorem B.128). $\square$

Lemma 3.7. If $f : [a, b] \rightarrow \mathbb{C}$ is integrable and satisfies $|f(x)| \leq C$ for all $x \in [a, b]$, then we have

$$\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx \leq (b-a)C. \quad (3.3)$$

Proof. The second inequality follows from the integral mean value theorem B.124. To prove the first inequality, define $\int_a^b f(x) dx = re^{i\theta}$ and we have

$$\begin{aligned} \left| \int_a^b f(x) dx \right| &= r = e^{-i\theta} \int_a^b f(x) dx = \int_a^b e^{-i\theta} f(x) dx \\ &= \operatorname{Re} \int_a^b e^{-i\theta} f(x) dx = \int_a^b \operatorname{Re} [e^{-i\theta} f(x)] dx \\ &\leq \int_a^b |e^{-i\theta} f(x)| dx = \int_a^b |f(x)| dx, \end{aligned}$$

where all steps follows from complex arithmetic. $\square$

Example 3.8 (Lemma 3.7 as a special case of the Minkowski inequality in L^p space). The first inequality in (3.3) is a direct consequence of the Minkowski inequality (Theorem F.11) on the L^p space,

$$\forall p \geq 1, \quad \|f + g\|_p \leq \|f\|_p + \|g\|_p, \quad (3.4)$$

where $f, g \in L^p(\Omega)$ and

$$\|f\|_p = \left(\int_{\Omega} |f(x)|^p dx \right)^{\frac{1}{p}}. \quad (3.5)$$

This inequality is important in establishing that $(L^p, \|\cdot\|_p)$ is a normed linear space for $p \geq 1$. For $p < 1$, (3.4) holds for non-negative functions f, g with the inequality reversed:

$$\forall p < 1, \quad \|f + g\|_p \geq \|f\|_p + \|g\|_p. \quad (3.6)$$

Then Lemma 3.7 follows from setting $f = u^2$, $g = v^2$, $p = \frac{1}{2}$ in (3.6) and using the inequality $\int_a^b u \leq \int_a^b |u|$.

Theorem 3.9. If $f : [a, b] \rightarrow \mathbb{R}^k$ is integrable, then $\|f\|$ is integrable and

$$\left\| \int_a^b f dx \right\| \leq \int_a^b \|f\| dx, \quad (3.7)$$

where $\|\cdot\|$ denotes the Euclidean norm in Definition C.176.

Proof. Write $f(x) := (f_1(x), \dots, f_k(x))$, $y_i := \int_a^b f_i(x) dx$, and we have $y := \int_a^b f(x) dx = (y_1, \dots, y_k)$ and

$$\|y\|^2 = \sum y_j^2 = \sum y_j \int_a^b f_j dx. \quad (3.8)$$

For a fixed $x \in [a, b]$, the Cauchy–Schwarz inequality yields

$$\langle y, f(x) \rangle = \sum y_i f_i(x) \leq \|y\| \|f(x)\|.$$

Integrate the above inequality, apply (3.8), and we have

$$\|y\|^2 = \sum y_j \int_a^b f_j dx \leq \|y\| \int_a^b \|f(x)\| dx.$$

If $\|y\| \neq 0$, we divide both sides of the above inequality by $\|y\|$ to obtain (3.7). If $\|y\| = 0$, each y_i must be zero, hence (3.7) also holds trivially. $\square$

Lemma 3.10. For continuously differentiable functions $u, v : [a, b] \rightarrow \mathbb{C}$, we have

$$\int_a^b u(x)v'(x)dx = (uv)\Big|_a^b - \int_a^b u'(x)v(x)dx. \quad (3.9)$$

Proof. This follows from Lemma 3.6 and the product rule $(uv)' = uv' + u'v$ in Theorem 2.20. $\square$

Lemma 3.11. Let $I_1, I_2 \subset \mathbb{R}$ be intervals, $a, b \in I_1$ and $\varphi : I_1 \rightarrow I_2$ continuously differentiable and $f : I_2 \rightarrow \mathbb{C}$ continuous. Then

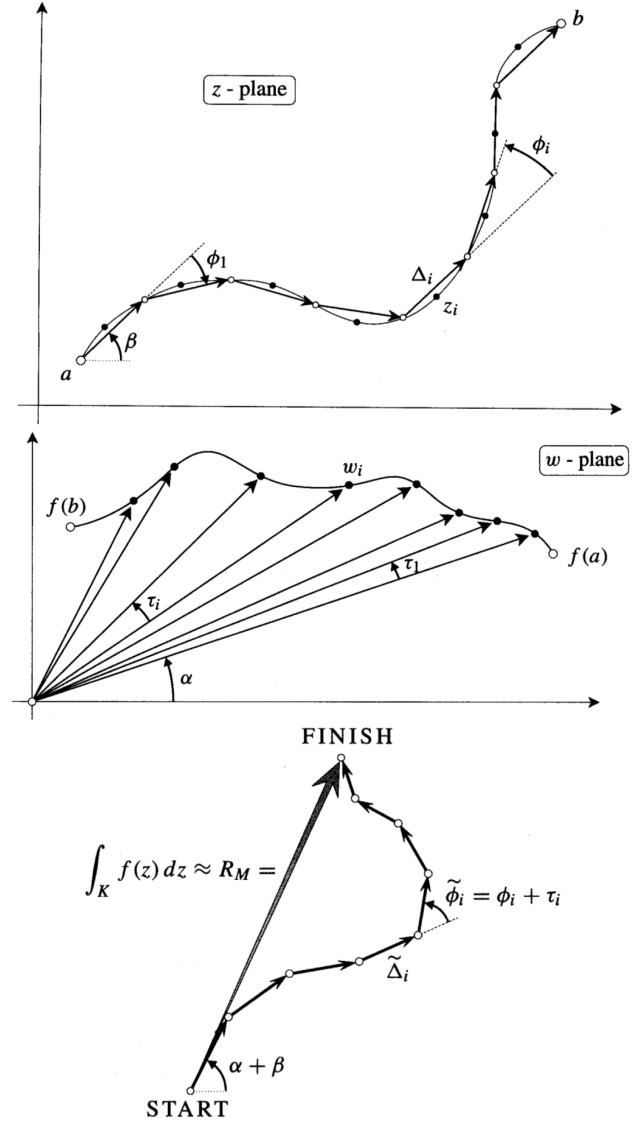
$$\int_{\varphi(a)}^{\varphi(b)} f(y)dy = \int_a^b f(\varphi(x))\varphi'(x)dx. \quad (3.10)$$

Proof. This follows directly from Theorem B.130 and Definition 3.4. $\square$

3.2 Complex line integrals

Definition 3.12. For a piecewise smooth curve $K : [a, b] \rightarrow \mathbb{C}$ and a continuous function $f : \mathcal{D} \rightarrow \mathbb{C}$ with $\mathcal{D} \supset K([a, b])$, the *complex line integral* or *complex contour integral* of f along K , written $\int_K f$, is given by

$$\int_K f(z)dz := \int_a^b f(K(t))K'(t)dt. \quad (3.11)$$



Lemma 3.13. Let $\alpha : [c, d] \rightarrow \mathbb{C}$ be a piecewise smooth curve and $f : \mathcal{D} \rightarrow \mathbb{C}$ be a continuous function with $\alpha([c, d]) \subset \mathcal{D} \subset \mathbb{C}$. Let $\varphi : [a, b] \rightarrow [c, d]$ be a continuously differentiable function with $\varphi(a) = c$ and $\varphi(b) = d$. Then we have

$$\int_{\alpha} f(\zeta)d\zeta = \int_{\alpha \circ \varphi} f(\zeta)d\zeta. \quad (3.12)$$

Proof. Definition 3.12 yields

$$\begin{aligned} \int_{\alpha} f(\zeta)d\zeta &= \int_c^d f(\alpha(t))\alpha'(t)dt \\ &= \int_a^b f(\alpha(\varphi(s)))\alpha'(\varphi(s))d\varphi(s) \\ &= \int_a^b f[(\alpha \circ \varphi)(s)](\alpha \circ \varphi)'(s)ds = \int_{\alpha \circ \varphi} f(\zeta)d\zeta. \quad \square \end{aligned}$$

Lemma 3.14. Let $f : \mathcal{D} \rightarrow \mathbb{C}$ be a continuous function with $\mathcal{D} \subset \mathbb{C}$ open. If f has a primitive F , i.e. $F' = f$, then for any piecewise smooth curve $\alpha \subset \mathcal{D}$ we have

$$\int_{\alpha} f(\zeta) d\zeta = F(\alpha(b)) - F(\alpha(a)). \quad (3.13)$$

Proof. Let $F = U + iV$, then the Cauchy–Riemann equations yield $f = \frac{\partial U}{\partial x} + i \frac{\partial V}{\partial x} = \frac{\partial V}{\partial y} - i \frac{\partial U}{\partial y}$. Write $\alpha(t) = x(t) + iy(t)$ and we have

$$\begin{aligned} f(\alpha(t))\alpha'(t) &= \frac{\partial U}{\partial x}x' + \frac{\partial U}{\partial y}y' + i \left(\frac{\partial V}{\partial x}x' + \frac{\partial V}{\partial y}y' \right) \\ &= \frac{dU}{dt} + i \frac{dV}{dt}. \end{aligned}$$

Then (3.13) follows from integrating the above identity over α and applying (3.11) and the fundamental theorem of calculus (Theorem B.128). $\square$

Corollary 3.15. $\forall z \in \mathbb{C}_-, \text{Log}(z) = L(z) := \int_1^z \frac{1}{\zeta} d\zeta$.

Proof. Corollary 2.26 states that $\text{Log}(z)$ is analytic in $\mathbb{C}_-$ and $(\text{Log}(z))' = \frac{1}{z}$. The rest follows from Lemma 3.14. $\square$

Theorem 3.16. Let $f : \mathcal{D} \rightarrow \mathbb{C}$ be a continuous function with $\mathcal{D} \subset \mathbb{C}$ open. If f has a primitive F , i.e. $F' = f$, then for any closed piecewise smooth curve γ in $\mathcal{D}$ we have

$$\oint_{\gamma} f(\zeta) d\zeta = 0. \quad (3.14)$$

Proof. Decompose γ into two curves $\alpha, \beta : [0, 1] \rightarrow \mathbb{C}$ such that $\gamma = \alpha \cup \beta$, $\alpha(0) = \beta(1)$, and $\alpha(1) = \beta(0)$. The proof is completed by $\oint_{\gamma} = \int_{\alpha} + \int_{\beta}$ and Lemma 3.14. $\square$

Corollary 3.17. For a counterclockwise oriented circle $\gamma(t) = r \exp(it)$ where $r > 0$ and $t \in [0, 2\pi]$, we have

$$\forall n \in \mathbb{Z}, \quad \oint_{\gamma} \zeta^n d\zeta = \begin{cases} 2\pi i & \text{if } n = -1, \\ 0 & \text{otherwise.} \end{cases} \quad (3.15)$$

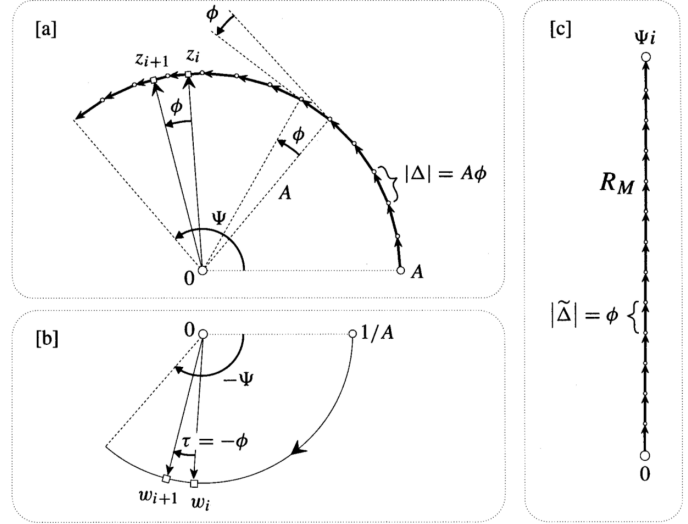
Proof. For $n \neq -1$, the function $f(z) = z^n$ has the primitive $F(z) = \frac{z^{n+1}}{n+1}$, c.f. Exercise 2.22. For $n = -1$,

$$\oint_{\gamma} \zeta^{-1} d\zeta = \int_0^{2\pi} (re^{it})^{-1} r i e^{it} dt = 2\pi i. \quad \square$$

Exercise 3.18 (An alternative proof of Corollary 3.17 via Corollary 3.15). By Corollary 2.26, we have $(\text{Log} z)' = \frac{1}{z}$ for $z \in \mathbb{C}_-$. By Definition 3.24, the unit circle $\mathbb{S}^1 = \alpha \oplus \beta$ is the concatenation of two curves $\theta \mapsto e^{i\theta}$,

$$\alpha := ((-\pi)^+, 0) \rightarrow \mathbb{C}, \quad \beta := (0, \pi) \rightarrow \mathbb{C}.$$

Put $z_{-1}^- := \alpha((-\pi)^+)$ and $z_{-1}^+ := \beta(\pi)$. Based on the above constructions, starting from the variable substitution $\zeta = rz$, calculate $\oint_{\gamma} \zeta^{-1} d\zeta$.



Corollary 3.19. The function $f(z) = \frac{1}{z}$ does not have a primitive if its domain is $\mathbb{C}^*$.

Proof. Otherwise it would contradict Corollary 3.17. $\square$

Exercise 3.20. Draw plots similar to those of $\int_K \frac{1}{z} dz$ to give a geometric interpretation of $\int_A^B z^m dz$ where $m \in \mathbb{N}^+$ and the integral is on the arc $\gamma(t) = r \exp(it)$ with $t \in [\text{Arg} A, \text{Arg} B]$. Then deduce the formula

$$\int_A^B z^m dz = \frac{1}{m+1} (B^{m+1} - A^{m+1}). \quad (3.16)$$

Exercise 3.21. Give a geometric interpretation of $\int_A^B e^z dz$ and the integral is on the vertical segment $\gamma(t) = A + it$ with $t \in [0, \theta]$, where $B = A + i\theta$ and $\theta > 0$. Then deduce the formula

$$\int_A^B e^z dz = e^B - e^A. \quad (3.17)$$

Exercise 3.22. In Corollary 3.17, we computed the integral $\oint_{\gamma} \zeta^n d\zeta$ for all integers n , where γ is a circle centered at the origin with positive orientation. Without invoking the Cauchy integral theorem, calculate $\oint_{\gamma} \zeta^n d\zeta$ where

- (a) γ is any circle not enclosing the origin;
- (b) γ is a positively oriented square with vertices at $\pm 1 \pm i$.

Hint: For each of the four line segments, use the parameterization and calculate the real integral $\int_{-1}^1 \frac{1}{1+t^2} dt$.

Exercise 3.23. For the counterclockwise oriented circle $\gamma(t) = r \exp(it)$ with $t \in [0, 2\pi]$ and $r \in (|a|, |b|)$, prove

$$\oint_{\gamma} \frac{1}{(\zeta - a)(\zeta - b)} d\zeta = \frac{2\pi i}{a - b}.$$

Hint: Express $\frac{1}{\zeta - a}$ as the geometric series expansion $\frac{1}{1 - z} = \sum_{n=0}^{\infty} z^n$ for $|z| < 1$.

3.3 The Cauchy integral theorem

3.3.1 Simply connected domains

Definition 3.24. The *concatenation of two curves* $\alpha : [a, b] \rightarrow \mathbb{C}$ and $\beta : [b, c] \rightarrow \mathbb{C}$ with $\alpha(b) = \beta(b)$ is another curve $\alpha \oplus \beta : [a, c] \rightarrow \mathbb{C}$ given by

$$(\alpha \oplus \beta)(t) := \begin{cases} \alpha(t) & \text{if } t \in [a, b], \\ \beta(t) & \text{if } t \in [b, c]. \end{cases} \quad (3.18)$$

Example 3.25. $\alpha(t) = \sqrt{1-t^2}$, $t \in [-1, 0]$ and $\beta(t) = 1$, $t \in [0, 2]$ are two smooth curves. The concatenation $\alpha \oplus \beta$ is not smooth but piecewise smooth: the second derivatives of α and β are different at $t = 0$. Indeed, the curvature of a unit circle and a straight line is 1 and 0, respectively.

Definition 3.26. The *reciprocal curve* of a curve $\alpha : [a, b] \rightarrow \mathbb{C}$ is the curve $\alpha^-(t) : [a, b] \rightarrow \mathbb{C}$ defined by $\alpha^-(t) = \alpha(b + a - t)$.

Definition 3.27. A set $\mathcal{D} \subset \mathbb{C}$ is called *arcwise connected* or *path connected* iff for any two points $z, w \in \mathcal{D}$ there is a piecewise smooth curve $\alpha : [a, b] \rightarrow \mathcal{D}$ such that $\alpha(a) = z$ and $\alpha(b) = w$. In particular, an arcwise connected nonempty open set is called a *domain*.

Example 3.28. The point set $\mathcal{D} = \{z \in \mathbb{C} : |z^2 - 3| < 1\}$ is not a domain because it is not connected: $\pm\sqrt{3} \in \mathcal{D}$ whereas $\{yi : y \in \mathbb{R}\} \cap \mathcal{D} = \emptyset$.

Example 3.29. The set $\mathcal{D} \setminus \mathcal{D}'$ is a domain where

$$\mathcal{D} = \{x + iy : x \in (0, 1), y \in (0, 1)\};$$

$$\mathcal{D}' = \cup_{n=2}^{\infty} \left\{ x + iy : x = \frac{1}{n}, y \in \left(0, \frac{1}{2}\right] \right\},$$

because $\mathcal{D} \setminus \mathcal{D}' = \mathcal{D} \setminus \overline{\mathcal{D}'} = \mathcal{D} \cap (\mathbb{C} \setminus \overline{\mathcal{D}'}): \overline{\mathcal{D}'}$ is closed, $(\mathbb{C} \setminus \overline{\mathcal{D}'})$ is open, and thus $\mathcal{D} \setminus \mathcal{D}'$ is open. $\mathcal{D} \setminus \mathcal{D}'$ is also path connected.

Definition 3.30. A *path* in a topological space $\mathcal{X}$ is a continuous map $\zeta : [0, 1] \rightarrow \mathcal{X}$.

Definition 3.31. A *loop* or *cycle* is a path whose endpoints coincide; this common endpoint is the *basepoint* of the loop.

Definition 3.32. A *homotopy of paths* in $\mathcal{X}$ is a family of paths $h_t : [0, 1] \rightarrow \mathcal{X}$, $t \in [0, 1]$, such that $h_t(0)$, $h_t(1)$ are independent of time and the associated map $H : [0, 1]^2 \rightarrow \mathcal{X}$ defined by $H(s, t) = h_t(s)$ is continuous. Two paths are *homotopic* if they are connected by a homotopy.

Example 3.33. The family of flow maps of a well-posed ODE $\frac{dx}{dt} = \mathbf{u}(\mathbf{x}, t)$ that act on a path at $t = 0$ is a homotopy of paths in the phase space of the ODE.

Lemma 3.34. The relation of homotopy on paths with fixed endpoints in any space is an equivalence relation.

Proof. This follows from Definitions 3.32 and A.36. $\square$

Definition 3.35. A *free homotopy* between two loops γ_1 and γ_2 in $\mathcal{X}$ is a function $H_f : [0, 1]^2 \rightarrow \mathcal{X}$ such that $H_f(s, 0) = \gamma_1(s)$ and $H_f(s, 1) = \gamma_2(s)$ for all s , and $H_f(0, t) = H_f(1, t)$ for all t . Then γ_1 and γ_2 are said to be *freely homotopic* in $\mathcal{X}$.

Example 3.36. If the basepoint of a free loop homotopy is further required to be fixed, i.e., $H_f(0, t) = H_f(1, t) = x_0$ for all $t \in [0, 1]$, we get a *based homotopy* or a *path homotopy*, in which case we also say that the two loops are freely homotopic relative to their endpoints.

Example 3.37. In the space of $\mathbb{S}^1 = [0, 1]/\{0, 1\}$, the loops $\alpha(s) := e^{2\pi i s}$ and $\beta(s) := 1$ are freely homotopic through $H_f(s, t) = e^{2\pi i s(1-t)}$. If the basepoint at $t = 0$ is chosen to be $x_*(0) = \alpha(1)$, then the expression of the basepoint at time t is $x_*(t) = e^{2\pi i(1-t)}$. During this homotopy, x_* covers $\mathbb{S}^1$ once. However, if x_* is required to be fixed at $\alpha(1)$, the loop α cannot be deformed into the constant loop β .

Definition 3.38. A topological space $\mathcal{X}$ is *simply connected* if and only if it is path connected and whenever $\alpha : [0, 1] \rightarrow \mathcal{X}$ and $\beta : [0, 1] \rightarrow \mathcal{X}$ are two paths with the same start and end points ($\alpha(0) = \beta(0)$ and $\alpha(1) = \beta(1)$), then there exists a based homotopy $H : [0, 1]^2 \rightarrow \mathcal{X}$ such that $H(s, 0) = \alpha(s)$ and $H(s, 1) = \beta(s)$, i.e., α can be continuously deformed into β while keeping both endpoints fixed.

Exercise 3.39. Prove that Theorem 2.73 holds not only for rectangular domains but also for simply connected domains.

3.3.2 A proof by Riemann

Theorem 3.40. For a continuous function $f : \mathcal{D} \rightarrow \mathbb{C}$ with $\mathcal{D}$ a domain, the following statements are equivalent:

- (a) f has a primitive.
- (b) The integral of f along any piecewise smooth closed curve in $\mathcal{D}$ vanishes.
- (c) The integral of f along any piecewise smooth curve in $\mathcal{D}$ depends only on the beginning and end points of the curve.

Proof. By Theorem 3.16, (a) implies (b).

To show (b) implies (c), let $\alpha : [a, b] \rightarrow \mathcal{D}$ and $\beta : [c, d] \rightarrow \mathcal{D}$ be two curves with the same starting and ending points. WLOG we can assume $b = c$. Otherwise set $\varphi(t) = t + c - b$ in Lemma 3.13 and we have $\int_{\beta} f(\zeta) d\zeta = \int_{\beta \circ \varphi} f(\zeta) d\zeta$. It follows that

$$0 = \oint_{\alpha \oplus \beta^-} f = \int_{\alpha} f - \int_{\beta} f.$$

To show (c) implies (a), fix a point $z_* \in \mathcal{D}$ and construct the integral

$$F(z) = \int_{z_*}^z f(\zeta) d\zeta, \quad (3.19)$$

where the integral is along a piecewise smooth curve connecting z_* with z in $\mathcal{D}$. It follows from Lemma 2.18 that $F'(z_0) = f(z_0)$ holds for any given $z_0 \in \mathcal{D}$ if the residual

$$r(z) := F(z) - F(z_0) - f(z_0)(z - z_0)$$

satisfies

$$\lim_{z \rightarrow z_0} \frac{r(z)}{|z - z_0|} = 0. \quad (3.20)$$

Since $z_0 \in \mathcal{D}$, there exists an open ball $\mathcal{U}_\rho(z_0)$ that is a subset of $\mathcal{D}$. For all $z \in \mathcal{U}_\rho(z_0)$, we have

$$F(z) = \int_{z_*}^{z_0} f(\zeta) d\zeta + \int_{z_0}^z f(\zeta) d\zeta = F(z_0) + \int_{z_0}^z f(\zeta) d\zeta,$$

where the last integral can be taken from z_0 to z along the line segment $\sigma : [0, 1] \rightarrow \mathcal{D}$ given by $\sigma(t) := z_0 + t(z - z_0)$. Definition 3.12 yields $\int_\sigma d\zeta = z - z_0$. Hence we have

$$r(z) = \int_\sigma (f(\zeta) - f(z_0)) d\zeta$$

and the continuity of f gives

$$\forall \epsilon > 0, \exists \delta \in (0, \rho), \text{ s.t. } |z - z_0| < \delta \Rightarrow |f(z) - f(z_0)| < \epsilon.$$

Then Lemma 3.7 yields (3.20). The proof is completed by z_0 being arbitrary. $\square$

Exercise 3.41. Determine whether the following functions have primitives on the indicated domains and fully justify your answer.

- (a) $f(z) = \frac{1}{z}$ on $\mathbb{C}^\bullet$;
- (b) $g(z) = \bar{z}$ on $\mathbb{C}$.

Exercise 3.42. Let $\gamma : [0, 1] \rightarrow \mathbb{H} := \{z \in \mathbb{C} : \text{Im } z > 0\}$ be a piecewise smooth curve with $\gamma(0) = 1 + \mathbf{i}$ and $\gamma(1) = \mathbf{i}$.

- (a) Do z and $\frac{1}{z}$ have primitives on $\mathbb{H}$? Justify your answer.
- (b) Calculate $\int_\gamma z dz$ and $\int_\gamma \frac{1}{z} dz$.

Theorem 3.43 (Green). Let α be a counter-clockwise oriented, piecewise smooth, simple closed curve in the plane. Denote by $\mathcal{D}$ the bounded complement of α and $u, v : \mathcal{D} \rightarrow \mathbb{R}$ functions with continuous partial derivatives on $\mathcal{D}$. Then

$$\oint u(x, y) dx + v(x, y) dy = \iint \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) dx dy. \quad (3.21)$$

Lemma 3.44. If $f(z) = u(x, y) + \mathbf{i}v(x, y)$ is continuous along a piecewise smooth curve α , then $f(z)$ is integrable along α and

$$\int_\alpha f(z) dz = \int_\alpha u dx - v dy + \mathbf{i} \int_\alpha v dx + u dy. \quad (3.22)$$

Proof. This follows from $dz = dx + \mathbf{i}dy$, Definition 3.4, and Theorem B.119 that a continuous function is integrable. $\square$

Theorem 3.45 (Cauchy). If $f : \mathcal{D} \rightarrow \mathbb{C}$ is analytic in a simply connected domain $\mathcal{D}$ and f' is continuous in $\mathcal{D}$, then

$$\oint_{\partial \mathcal{D}} f(\zeta) d\zeta = 0. \quad (3.23)$$

Proof. Let $f(z) = u(x, y) + \mathbf{i}v(x, y)$ and $z = x + \mathbf{i}y$. In 1851, Riemann reasoned that

$$\begin{aligned} \oint_{\partial \mathcal{D}} f(\zeta) d\zeta &= \oint_{\partial \mathcal{D}} u dx - v dy + \mathbf{i} \oint_{\partial \mathcal{D}} v dx + u dy \\ &= \iint_{\mathcal{D}} -\left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) dx dy + \mathbf{i} \iint_{\mathcal{D}} \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right) dx dy = 0, \end{aligned}$$

where the first step follows from Lemma 3.44, the second from Theorem 3.43, and the third from Theorem 2.50. $\square$

Lemma 3.46 (Polynomial growth). For a complex polynomial of degree $n \in \mathbb{N}^+$,

$$P(z) = \sum_{j=0}^n a_j z^j, \quad (3.24)$$

where $a_j \in \mathbb{C}$ for all $j = 0, \dots, n$ and $a_n \neq 0$, we have

$$\frac{1}{2}|a_n||z|^n \leq |P(z)| \leq \frac{3}{2}|a_n||z|^n \quad (3.25)$$

for any z with a sufficiently large modulus $|z| \geq \rho$, e.g.,

$$\rho := \max \left\{ 1, \frac{2}{|a_n|} \sum_{j=0}^{n-1} |a_j| \right\}. \quad (3.26)$$

Proof. Define $Q(z) = P(z) - a_n z^n$. Then the triangle inequality and the condition $|z| \geq \rho$ yield

$$|Q(z)| \leq \left(\sum_{j=0}^{n-1} |a_j| \right) |z|^{n-1} \leq \frac{|a_n|}{2} |z|^n.$$

Apply the triangle inequality, i.e. Corollary 1.66, again to $Q(z) + a_n z^n = P(z)$ and we have

$$|a_n||z|^n - |Q(z)| \leq |P(z)| \leq |a_n||z|^n + |Q(z)|.$$

Combining the two inequalities yields (3.25). $\square$

Theorem 3.47 (Fundamental theorem of Algebra). Any nonconstant complex polynomial (3.24) has a root.

Proof. Suppose $P(z) \neq 0$ for all $z \in \mathbb{C}$. Define $Q(z)$ by $P(z) = zQ(z) + a_0$. Then for any $z \neq 0$ we have

$$\frac{1}{z} = \frac{zQ(z) + a_0}{zP(z)} = \frac{Q(z)}{P(z)} + \frac{a_0}{zP(z)}.$$

By Corollary 3.17 and Theorem 3.45, the integral of $\frac{1}{z}$ along the circle $\alpha(t) = R \exp(\mathbf{i}t)$ with $t \in [0, 2\pi]$ and $R > 0$ yields

$$2\pi \mathbf{i} = \oint_\alpha \frac{a_0}{zP(z)} dz,$$

where we have used the facts of $P(z) \neq 0$ and P/Q being analytic. Then Lemma 3.7 and Lemma 3.46 yield

$$2\pi = \left| \oint_\alpha \frac{a_0}{zP(z)} dz \right| \leq \oint_\alpha \left| \frac{a_0}{zP(z)} \right| dz \leq 2\pi R \frac{2|a_0|}{R|a_n|R^n},$$

which cannot hold for any sufficiently large $R > 0$. $\square$

3.3.3 A proof by Goursat

Definition 3.48. The triangle spanned by $z_1, z_2, z_3 \in \mathbb{C}$ is the point set

$$\Delta := \left\{ \sum_{j=1}^3 t_j z_j : t_j \geq 0, \sum_{j=1}^3 t_j = 1 \right\}. \quad (3.27)$$

The triangular path $\langle z_1, z_2, z_3 \rangle$ is the closed curve

$$\langle z_1, z_2, z_3 \rangle = \alpha := \alpha_1 \oplus \alpha_2 \oplus \alpha_3, \quad (3.28)$$

where the edges constituting the triangle boundary are

$$\alpha_1(t) = z_1 + (t-0)(z_2 - z_1), \quad t \in [0, 1];$$

$$\alpha_2(t) = z_2 + (t-1)(z_3 - z_2), \quad t \in [1, 2];$$

$$\alpha_3(t) = z_3 + (t-2)(z_1 - z_3), \quad t \in [2, 3].$$

Lemma 3.49. If $\{\mathcal{P}_i\}$ is a collection of compact subsets of a metric space such that the intersection of every finite subcollection of $\{\mathcal{P}_i\}$ is nonempty, then $\cap_i \mathcal{P}_i$ is nonempty.

Proof. Denote by $\mathcal{Q}_i$ the complement of $\mathcal{P}_i$. Suppose there exists $\mathcal{P}_j$ such that no points of $\mathcal{P}_j$ belongs to every $\mathcal{P}_i$. The the sets $\{\mathcal{Q}_i\}$ is an open cover of $\mathcal{P}_j$. By Definition D.145, there exist finitely many indices $k_1, \dots, k_n$ such that $\mathcal{P}_j \subset \cup_{\ell=k_1}^{k_n} \mathcal{Q}_\ell$. But this implies $\mathcal{P}_j \cap \mathcal{P}_{k_1} \cap \mathcal{P}_{k_2} \cap \dots \cap \mathcal{P}_\ell = \emptyset$, contradicting the starting point. $\square$

Theorem 3.50 (Cantor intersection theorem). Suppose $(\mathcal{P}_k)$ is a sequence of non-empty, compact subsets of $\mathcal{P}$ satisfying $\mathcal{P}_0 \supseteq \mathcal{P}_1 \supseteq \dots \supseteq \mathcal{P}_k \supseteq \dots$. Then $\cap_k \mathcal{P}_k \neq \emptyset$.

Proof. This follows directly from Lemma 3.49. $\square$

Exercise 3.51. Prove that if $\{E_n\}$ is a sequence of closed nonempty and bounded sets in a complete metric space $\mathcal{X}$, if $E_{n+1} \subset E_n$, and if $\lim_{n \rightarrow \infty} \text{diam } E_n = 0$, cf. Definition D.22. Then $\cap_{n=1}^{\infty} E_n$ consists of exactly one point.

Exercise 3.52. Use the conclusion in Exercise 3.51 to prove the Baire category theorem D.80.

Theorem 3.53 (Cauchy integral theorem on triangular paths). If $f : \mathcal{D} \rightarrow \mathbb{C}$ is analytic on a domain $\mathcal{D}$ and the triangle $\Delta(z_1, z_2, z_3)$ is a subset of $\mathcal{D}$, then we have

$$\oint_{\langle z_1, z_2, z_3 \rangle} f(\zeta) d\zeta = 0. \quad (3.29)$$

Proof. As shown in the plot below, we first construct a sequence of triangular paths

$$\alpha^{(n)} = \left\langle z_1^{(n)}, z_2^{(n)}, z_3^{(n)} \right\rangle$$

by inductively defining

$$\alpha^{(0)} := \alpha = \langle z_1, z_2, z_3 \rangle,$$

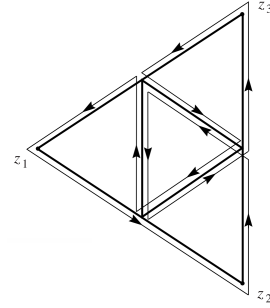
$$\alpha_1^{(n)} := \left\langle \frac{z_1^{(n)} + z_2^{(n)}}{2}, z_2^{(n)}, \frac{z_2^{(n)} + z_3^{(n)}}{2} \right\rangle,$$

$$\alpha_2^{(n)} := \left\langle \frac{z_2^{(n)} + z_3^{(n)}}{2}, z_3^{(n)}, \frac{z_3^{(n)} + z_1^{(n)}}{2} \right\rangle,$$

$$\alpha_3^{(n)} := \left\langle \frac{z_3^{(n)} + z_1^{(n)}}{2}, z_1^{(n)}, \frac{z_1^{(n)} + z_2^{(n)}}{2} \right\rangle,$$

$$\alpha_4^{(n)} := \left\langle \frac{z_1^{(n)} + z_2^{(n)}}{2}, \frac{z_2^{(n)} + z_3^{(n)}}{2}, \frac{z_3^{(n)} + z_1^{(n)}}{2} \right\rangle.$$

and choosing $\alpha^{(n+1)} = \alpha_1^{(n)}$ or $\alpha_2^{(n)}$ or $\alpha_3^{(n)}$ or $\alpha_4^{(n)}$.



It follows that

$$\oint_{\alpha^{(n)}} f = \oint_{\alpha_1^{(n)}} f + \oint_{\alpha_2^{(n)}} f + \oint_{\alpha_3^{(n)}} f + \oint_{\alpha_4^{(n)}} f$$

and we can choose $\alpha^{(n+1)}$ such that

$$\left| \oint_{\alpha^{(n)}} f \right| \leq 4 \left| \oint_{\alpha^{(n+1)}} f \right|,$$

which implies

$$\left| \oint_{\alpha} f \right| \leq 4^n \left| \oint_{\alpha^{(n)}} f \right|.$$

The closed triangles $\Delta^{(n)}$ are nested and Exercise 3.51 states that there exists a point $z_0 \in \cap_n \Delta^{(n)} \subset \mathcal{D}$. By Lemma 2.18, the condition of f being analytic at z_0 yields $f(z) = f(z_0) + f'(z_0)(z - z_0) + r(z)$ where

$$\lim_{z \rightarrow z_0} \frac{r(z)}{|z - z_0|} = 0. \quad (3.30)$$

Then we have $\oint_{\alpha^{(n)}} f(\zeta) d\zeta = \oint_{\alpha^{(n)}} r(\zeta) d\zeta$ from Theorem 3.16 and Exercise 2.22. It follows that

$$\left| \oint_{\alpha} f(\zeta) d\zeta \right| \leq 4^n \left| \oint_{\alpha^{(n)}} r(\zeta) d\zeta \right|. \quad (3.31)$$

We show $\lim_{n \rightarrow \infty} 4^n \left| \oint_{\alpha^{(n)}} r(\zeta) d\zeta \right| = 0$ to complete the proof. By (3.30), we have

$$\forall \epsilon > 0, \exists \delta > 0 \text{ s.t. } |z - z_0| < \delta \Rightarrow |r(z)| \leq \epsilon |z - z_0|, \quad (3.32)$$

where the condition $|z - z_0| < \delta$ holds for any sufficiently large n . In other words, we can choose n large enough such that $\Delta^{(n)} \subset \mathcal{U}_\delta(z_0)$. Denote by $\ell(\alpha)$ the arc length of α as

in Definition C.382. Then by construction of the sequence of triangles, we have

$$|z - z_0| \leq \ell(\alpha^{(n)}) = \frac{1}{2^n} \ell(\alpha),$$

which, together with (3.31), (3.32), and Lemma 3.7, yields

$$\left| \oint_{\alpha} f(\zeta) d\zeta \right| \leq 4^n \epsilon \frac{1}{2^n} \ell(\alpha) \ell(\alpha^{(n)}) = \epsilon \ell^2(\alpha).$$

To sum up, $\left| \oint_{\alpha} f(\zeta) d\zeta \right|$ can be made arbitrarily small, hence (3.29) holds. $\square$

Corollary 3.54. If $f : \mathcal{D} \rightarrow \mathbb{C}$ is analytic in a simply connected domain $\mathcal{D}$, then for any piecewise linear simple closed curve $\beta \subset \mathcal{D}$ we have

$$\oint_{\beta} f(\zeta) d\zeta = 0. \quad (3.33)$$

Proof. This follows from Theorem 3.53 and the two ears theorem, which states that every simple polygon with more than three vertices has at least two ears. $\square$

Theorem 3.55 (Heine 1872). If $K \subset \mathbb{C}$ is compact and $f : K \rightarrow \mathbb{C}$ is continuous, then f is uniformly continuous.

Proof. This is a specialization of Theorem D.150. $\square$

Lemma 3.56. If $f : \mathcal{D} \rightarrow \mathbb{C}$ is continuous in a domain $\mathcal{D}$, then for any piecewise smooth simple curve $\alpha \subset \mathcal{D}$ and any $\epsilon > 0$, there exists a piecewise linear curve $\sigma \subset \mathcal{D}$ such that

$$\left| \int_{\alpha} f(\zeta) d\zeta - \int_{\sigma} f(\zeta) d\zeta \right| < \epsilon. \quad (3.34)$$

In other words, the integral of f on a piecewise smooth curve can be approximated arbitrarily well by an integral of f on line segments.

Proof. Since $\mathcal{D}$ is open, there exists a compact subset $G \subset \mathcal{D}$ such that $\alpha \cap \partial G = \emptyset$ and $\alpha \subset G$. Define

$$\rho = \inf\{|z_1 - z_2| : z_1 \in \alpha, z_2 \in \partial G\}.$$

Clearly $\rho > 0$. We have $\forall z \in \alpha, \forall r < \rho, \mathcal{U}_r(z) \subset G$ and

$$\forall z_1, z_2 \in \alpha, |z_1 - z_2| < \rho \Rightarrow \{z_1 t + z_2(1-t) : t \in [0, 1]\} \subset G.$$

By Theorem 3.55, the continuity of f and the compactness of G yield the uniform continuity of f on G , i.e.,

$$\forall \epsilon > 0, \exists \delta_1 > 0 \text{ s.t. } \forall z_1, z_2 \in G, \\ |z_1 - z_2| < \delta_1 \Rightarrow |f(z_1) - f(z_2)| < \frac{\epsilon}{2\ell}, \quad (3.35)$$

where ℓ is the arc length of α .

Choose $n + 1$ points $z_1, z_2, \dots, z_{n+1} \in \alpha$ such that $\max_{j=1}^n |z_j - z_{j+1}| < \min(\rho, \delta_1)$: if α is closed then we set $z_{n+1} = z_1$; if α is not closed we set z_1 and z_{n+1} as the starting and ending points, respectively. We also require that $\alpha^{-1}(z_1), \alpha^{-1}(z_2), \dots, \alpha^{-1}(z_{n+1})$ increases monotonically. Then the piecewise linear closed curve σ formed by connecting z_j 's consecutively by linear segments is contained

in $\mathcal{D}$. Its edges σ_j 's correspond to the arcs α_j 's. For each j , Lemma 3.14 yields

$$\int_{\sigma_j} f(z_j) d\zeta = f(z_j)(z_{j+1} - z_j) = \int_{\alpha_j} f(z_j) d\zeta. \quad (3.36)$$

Consequently, we have

$$\begin{aligned} & \left| \int_{\alpha} f(\zeta) d\zeta - \int_{\sigma} f(\zeta) d\zeta \right| \\ & \leq \left| \int_{\alpha_j} [f(\zeta) - f(z_j)] d\zeta \right| + \left| \int_{\sigma_j} [f(\zeta) - f(z_j)] d\zeta \right| \\ & \leq \ell(\alpha_j) \sup_{\zeta \in \alpha_j} |f(\zeta) - f(z_j)| + \ell(\sigma_j) \sup_{\zeta \in \sigma_j} |f(\zeta) - f(z_j)| \\ & < \frac{\epsilon}{2\ell} [\ell(\alpha_j) + \ell(\sigma_j)] \leq \frac{\epsilon}{\ell} \ell(\alpha_j), \end{aligned}$$

where the first step follows from (3.36) and the triangle inequality, the second step from Lemma 3.7 and Definition C.382, and the third step from (3.35). Then we have

$$\begin{aligned} & \left| \int_{\alpha} f(\zeta) d\zeta - \int_{\sigma} f(\zeta) d\zeta \right| = \left| \sum_{j=1}^n \int_{\alpha_j} f(\zeta) d\zeta - \int_{\sigma_j} f(\zeta) d\zeta \right| \\ & \leq \sum_{j=1}^n \left| \int_{\alpha_j} f(\zeta) d\zeta - \int_{\sigma_j} f(\zeta) d\zeta \right| = \epsilon, \end{aligned}$$

which completes the proof. $\square$

Theorem 3.57 (Cauchy integral theorem for simply connected domains). If $f : \mathcal{D} \rightarrow \mathbb{C}$ is analytic in a simply connected domain $\mathcal{D}$, then for any piecewise smooth closed curve $\alpha \subset \mathcal{D}$ we have

$$\oint_{\alpha} f(\zeta) d\zeta = 0. \quad (3.37)$$

Proof. This follows directly from Corollary 3.54 and Lemma 3.56. $\square$

Exercise 3.58. For an analytic function $f : \mathbb{C} \rightarrow \mathbb{C}$ satisfying $\forall z \in \mathbb{C}, f(z+1) = f(z)$, prove that the value of $\int_{[w, w+1]} f(\zeta) d\zeta$ is independent of the choice of $w \in \mathbb{C}$.

Exercise 3.59. For bounded simply connected domain $\mathcal{D}$,

- (a) prove $\oint_{\partial \mathcal{D}} \bar{z} dz = 2iS(\mathcal{D})$, where $S(\mathcal{D})$ is the area of $\mathcal{D}$,
- (b) show that f being analytic on $\bar{\mathcal{D}}$ implies

$$\max_{z \in \partial \mathcal{D}} |\bar{z} - f(z)| \geq 2 \frac{S(\mathcal{D})}{\ell(\partial \mathcal{D})},$$

where $\partial \mathcal{D}$ is the boundary of $\mathcal{D}$ and $\ell(\partial \mathcal{D})$ its arc length.

Definition 3.60. A *star-shaped domain* is an open set $\mathcal{D} \subset \mathbb{C}$ that satisfies

$$\exists z_* \in \mathcal{D} \text{ s.t. } \forall z \in \mathcal{D}, \{z_* + t(z - z_*) : t \in [0, 1]\} \subset \mathcal{D},$$

where z_* is called a *star center*.

Example 3.61. Each convex domain, in particular the open disk, is star-shaped. The ring sector

$$\{z_0 + \zeta \rho e^{i\varphi} : \rho \in (r, R), \varphi \in (0, \beta); \zeta, z_0 \in \mathbb{C}, |\zeta| = 1\}$$

is star-shaped if $\beta < \pi$ and $\cos \frac{\beta}{2} > \frac{r}{R}$.

Theorem 3.62 (Cauchy integral theorem for star-shaped domains). Each analytic function f defined on a star-shaped domain $\mathcal{D}$ has a primitive in $\mathcal{D}$.

Proof. This follows from Theorems 3.57 and 3.40 since a star-shaped domain is a simply connected domain. $\square$

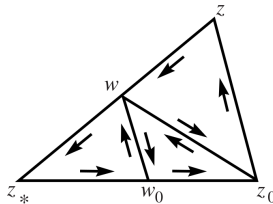
Exercise 3.63 (Alternative proof of Corollary 3.15). Prove Corollary 3.15 by Theorems 3.62 and 3.40.

Corollary 3.64. Let $f : \mathcal{D} \rightarrow \mathbb{C}$ be continuous in a star-shaped domain $\mathcal{D}$ with center z_* . If f is analytic at every point $z \neq z_*$, then f has a primitive in $\mathcal{D}$.

Proof. By Theorem 3.40, it suffices to show

$$\oint_{\langle z, z_0, z_* \rangle} f(\zeta) d\zeta = 0,$$

where $z \neq z_*$ and $z_0 \neq z_*$ are two arbitrary points in $\mathcal{D}$ and the triangle is entirely contained within $\mathcal{D}$.



As shown in the above plot, Theorem 3.53 yields

$$\oint_{\langle z_*, z_0, z \rangle} = \oint_{\langle z_*, w_0, w \rangle} + \oint_{\langle w_0, z_0, w \rangle} + \oint_{\langle z_0, z, w \rangle} = \oint_{\langle z_*, w_0, w \rangle}.$$

The proof is completed by passing to the limit $w \rightarrow z_*$ and $w_0 \rightarrow z_*$. $\square$

3.3.4 Multiply connected domains

Theorem 3.65 (Jordan curve theorem). The complement of a Jordan curve γ consists of two components: one is bounded and the other is unbounded; both of them are open and each has γ as its boundary.

Definition 3.66. A Jordan curve is said to be *positively oriented* or *counterclockwise oriented* iff an observer traversing in the direction of its parametrization always has the bounded complement on her left; it is *negatively oriented* or *clockwise oriented* iff the observer always has the unbounded complement on her left.

Definition 3.67. The *interior* of an oriented Jordan curve γ , denoted by $\text{int}(\gamma)$, is the complement of γ that always lies to the left of an observer who traverses the curve in the increasing direction of its parameter; the other complement of γ is called the *exterior* of γ and denoted by $\text{ext}(\gamma)$,

Lemma 3.68 (Seifert decomposition). Any oriented closed curve can be decomposed into the union of a finite number of oriented, pairwise non-crossing Jordan curves.

Proof. The local neighborhood of each self-intersection contains an even number of arc pieces. Rearrange them so that the resulting Jordan curves have no proper intersections. $\square$

Definition 3.69. A *disk with n holes* ($n \in \mathbb{N}$) is a connected open set $\mathcal{D} \subset \mathbb{C}$ that can be expressed as

$$\mathcal{D} := \bigcap_{j=0}^n \text{int}(\gamma_j), \quad (3.38)$$

where $\text{int}(\gamma)$ is the interior of a Jordan curve γ in Definition 3.67, γ_0 is a counter-clockwise oriented Jordan curve, and $\gamma_1, \dots, \gamma_n$ are clockwise oriented Jordan curves satisfying

$$\forall j = 1, \dots, n, \text{ext}(\gamma_j) \subset \text{int}(\gamma_0), \quad (3.39a)$$

$$\forall j, k = 1, \dots, n, j \neq k \Rightarrow \text{ext}(\gamma_j) \cap \text{ext}(\gamma_k) = \emptyset. \quad (3.39b)$$

The boundary and closure of a disk with holes are denoted respectively by

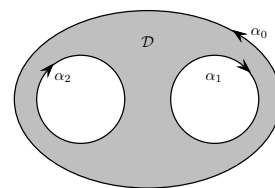
$$\partial\mathcal{D} := \bigcup_{j=0}^n \gamma_j, \quad \overline{\mathcal{D}} := \mathcal{D} \cup \partial\mathcal{D}.$$

Example 3.70. It can be shown that any regular open semianalytic set with bounded boundary is a disk with holes. For example, an open disk is a disk with no holes.

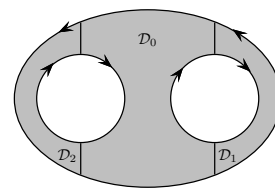
Theorem 3.71 (Cauchy integral theorem for a disk with holes). Let $\mathcal{D} := \bigcap_{j=0}^n \text{int}(\gamma_j)$ be a disk with holes as in Definition 3.69. If a complex function f is analytic on $\mathcal{D}$ and continuous on $\overline{\mathcal{D}} := \mathcal{D} \cup \partial\mathcal{D}$, then we have

$$\int_{\partial\mathcal{D}} f(z) dz = 0.$$

Proof. The domain $\mathcal{D}$ is exemplified for $n = 2$ as follows.



It follows from a simple induction that $\bigcup_{j=0}^n \gamma_j$ can be reorganized into positively-oriented Jordan curves with disjoint interiors $\mathcal{D}_0, \mathcal{D}_1, \dots, \mathcal{D}_m$ as follows.



The proof is then completed by $\bigcup_{j=0}^n \gamma_j = \bigcup_{j=0}^m \partial\mathcal{D}_j$ and the Cauchy integral theorem 3.57. $\square$

3.4 Cauchy integral formulas and some consequences

3.4.1 Cauchy integral formulas

Lemma 3.72. Define a counter-clockwise oriented circle as

$$\alpha(t) = c + re^{it}, \quad c \in \mathbb{C}, t \in [0, 2\pi], r > 0. \quad (3.40)$$

Then we have

$$\forall z \in \mathbb{C}, |z - c| < r \Rightarrow \oint_{\alpha} \frac{d\zeta}{\zeta - z} = 2\pi i. \quad (3.41)$$

Proof. Referring to Definition 3.69, we construct a disk $\mathcal{D}$ with one hole by setting γ_0 as the circle α and γ_1 as a clockwise oriented circle of which the center is z and the radius is $\rho \leq r - |c - z|$. Theorem 3.71 yields

$$0 = \int_{\partial\mathcal{D}} = \oint_{\gamma} + \oint_{\gamma_1} \Rightarrow \oint_{\gamma} = \oint_{\gamma_1^-}.$$

The rest of the proof follows from Corollary 3.17. $\square$

Theorem 3.73 (Cauchy integral formula). Let $\mathcal{D} \subseteq \mathbb{C}$ be open and $f : \mathcal{D} \rightarrow \mathbb{C}$ be an analytic function. Suppose that the closed disk $\overline{\mathcal{U}_r(c)}$ is completely contained in $\mathcal{D}$. Then

$$\forall z \in \mathcal{U}_r(c), \quad f(z) = \frac{1}{2\pi i} \oint_{\alpha} \frac{f(\zeta)}{\zeta - z} d\zeta, \quad (3.42)$$

where α is the circle given by (3.40).

Proof. By the compactness of $\overline{\mathcal{U}_r(c)}$, there exists an $R > r$ such that

$$\overline{\mathcal{U}_r(c)} \subset \mathcal{U}_R(c) \subset \mathcal{D}.$$

We can thus assume that $\mathcal{D}$ is an open disk. The function

$$g(w) := \begin{cases} \frac{f(w) - f(z)}{w - z} & \text{if } w \neq z, \\ f'(z) & \text{if } w = z \end{cases}$$

is analytic in $\mathcal{D} \setminus \{z\}$ and is continuous in $\mathcal{D}$. Then Corollary 3.64 and Theorem 3.40 yield

$$\oint \frac{f(\zeta) - f(z)}{\zeta - z} d\zeta = 0 \Rightarrow f(z) \oint \frac{1}{\zeta - z} d\zeta = \oint \frac{f(\zeta)}{\zeta - z} d\zeta.$$

The proof is completed by Lemma 3.72. $\square$

Corollary 3.74 (Cauchy integral formula for non-simply connected domains). Let γ_0 be a counter-clockwise oriented Jordan curve and $\gamma_1, \dots, \gamma_n$ be clockwise oriented Jordan curves satisfying (3.39). If a complex function f is analytic on the disk with holes $\mathcal{D} := \cap_{j=0}^n \text{int}(\gamma_j)$ and continuous on $\overline{\mathcal{D}}$, then we have

$$\oint_{\gamma_0} f(z) dz = \sum_{j=1}^n \oint_{\gamma_j^-} f(z) dz, \quad (3.43)$$

where each γ_j^- is γ_j with its orientation reversed, cf. Definition 3.26.

Proof. This follows from Theorem 3.71 by moving the summation term in (3.43) to the right-hand side (RHS) and reversing the orientation of γ_j 's for each $j = 1, 2, \dots, n$. $\square$

Example 3.75. Let α denote the circle defined in (3.40) with $c = 0$ and $r = 2$. We compute

$$\oint_{\alpha} \frac{\zeta}{(9 - \zeta^2)(\zeta + i)} d\zeta = \oint_{\alpha} \frac{\frac{\zeta}{9 - \zeta^2}}{\zeta - (-i)} d\zeta = 2\pi i \left. \frac{\zeta}{9 - \zeta^2} \right|_{\zeta = -i} = \frac{\pi}{5},$$

where the second step follows from Theorem 3.73.

Corollary 3.76 (Mean value equation). For the special case of $z = c$, Theorem 3.73 reduces to

$$f(c) = \frac{1}{2\pi} \int_0^{2\pi} f(c + re^{it}) dt. \quad (3.44)$$

Proof. This follows directly from Theorem 3.73 by substituting $z = c$ and $\zeta = c + re^{it}$. $\square$

Example 3.77. To compute $I = \oint_{|z|=2} \frac{dz}{z^2 + 2}$, we construct two counter-clockwise oriented circles α_1 and α_2 surrounding the two singularities $z = \pm\sqrt{2}i$. For sufficiently small radii, Corollary 3.74 yields

$$\begin{aligned} \oint_{|z|=2} f(z) dz &= \oint_{\alpha_1} f(z) dz + \oint_{\alpha_2} f(z) dz \\ &= \oint_{\alpha_1} \frac{1}{z - \sqrt{2}i} dz + \oint_{\alpha_2} \frac{1}{z - (-\sqrt{2}i)} dz \\ &= \left. \frac{2\pi i}{z - \sqrt{2}i} \right|_{z=\sqrt{2}i} + \left. \frac{2\pi i}{z - (-\sqrt{2}i)} \right|_{z=-\sqrt{2}i} = 0. \end{aligned}$$

Exercise 3.78. Compute $I = \oint_{|z|=1} \frac{3z-1}{z(z-2)} dz$.

Exercise 3.79. For some $r > 0$, f is an analytic function on $\{z \in \mathbb{C} : |z| > r\}$. Show that $\lim_{z \rightarrow \infty} zf(z) = A$ implies

$$\forall R > r, \quad \oint_{|z|=R} f(z) dz = 2\pi i A.$$

Exercise 3.80. For $R > 0$, write $\mathcal{U}_R := \{z \in \mathbb{C} : |z| < R\}$.

(a) Let $f : \mathcal{U}_{R_0} \rightarrow \mathbb{C}$ be an analytic function, where $R_0 > 0$. Prove that whenever $R \in (0, R_0)$ and $z \in \mathcal{U}_R$, then

$$f(z) = \frac{1}{2\pi} \int_0^{2\pi} f(Re^{it}) \frac{R^2 - |z|^2}{|Re^{it} - z|^2} dt.$$

(b) Show that

$$\text{Re} \left(\frac{Re^{it} + z}{Re^{it} - z} \right) = \frac{R^2 - |z|^2}{|Re^{it} - z|^2}.$$

(c) If u is harmonic in the unit disk $\mathbb{D} = \mathcal{U}_1$ and continuous on $\overline{\mathbb{D}}$, then show that for $z \in \mathbb{D}$,

$$u(z) = \int_0^{2\pi} P(z, e^{it}) u(e^{it}) dt,$$

where $P(z, e^{it})$ is the *Poisson kernel* defined by

$$P(z, e^{it}) = \frac{1}{2\pi} \frac{1 - |z|^2}{|e^{it} - z|^2} = \frac{1}{2\pi} \text{Re} \left(\frac{e^{it} + z}{e^{it} - z} \right).$$

Hint: For (a), note that if $w = \frac{R^2}{\bar{z}}$, then the integral of $\frac{f(\zeta)}{\zeta - w}$ over the circle $|\zeta| = R$ is zero. Use this, together with the usual Cauchy integral formula, to deduce the desired result. For (c), apply Exercise 3.39, then use (a) and (b).

Theorem 3.81 (Leibniz integral rule). Let $f : [a, b] \times [c, d] \rightarrow \mathbb{R}$ be a continuous function. Suppose that the partial derivative $\frac{\partial f(t, x)}{\partial x}$ exists and is continuous. Also suppose that the functions $\alpha, \beta : [c, d] \rightarrow [a, b]$ are both continuously differentiable. Then

$$\begin{aligned} & \frac{d}{dx} \left(\int_{\alpha(x)}^{\beta(x)} f(t, x) dt \right) \\ &= \int_{\alpha(x)}^{\beta(x)} \frac{\partial f(t, x)}{\partial x} dt + f(\beta(x), x) \frac{d\beta(x)}{dx} - f(\alpha(x), x) \frac{d\alpha(x)}{dx}. \end{aligned} \quad (3.45)$$

Proof. For $F(\alpha, \beta, x) := \int_{\alpha(x)}^{\beta(x)} f(t, x) dt$, the chain rule gives

$$\frac{d}{dx} F(\alpha(x), \beta(x), x) = \frac{\partial F}{\partial \alpha} \frac{d\alpha(x)}{dx} + \frac{\partial F}{\partial \beta} \frac{d\beta(x)}{dx} + \frac{\partial F}{\partial x}.$$

The first fundamental theorem of calculus (Theorem B.127) yields $\frac{\partial F}{\partial \alpha} = -f(\alpha, x)$ and $\frac{\partial F}{\partial \beta} = f(\beta, x)$. Then (3.45) is given by $\frac{\partial F}{\partial x} = \int_{\alpha(x)}^{\beta(x)} \frac{\partial f(t, x)}{\partial x} dt$, which holds because

$$\begin{aligned} & \int_u^v \frac{\partial f}{\partial x}(t, x) dt = \frac{d}{dx} \int_u^x \int_u^v \frac{\partial f}{\partial y}(t, y) dy dt \\ &= \frac{d}{dx} \int_u^x \int_c^x \frac{\partial f}{\partial y}(t, y) dy dt = \frac{d}{dx} \int_u^x [f(t, x) - f(t, c)] dt \\ &= \frac{d}{dx} \int_u^x f(t, x) dt, \end{aligned}$$

where the first step follows from Theorem B.127, the second from Fubini's theorem and the condition of $\frac{\partial f(t, x)}{\partial x}$ being continuous, and third from the second fundamental theorem of calculus (Theorem B.128). $\square$

Lemma 3.82. Let $\mathcal{D} \subseteq \mathbb{C}$ be open and $f : [a, b] \times \mathcal{D} \rightarrow \mathbb{C}$ be a continuous function that is also analytic in $\mathcal{D}$ for any fixed $t \in [a, b]$. Also suppose that the partial derivative $\frac{\partial f}{\partial z} : [a, b] \times \mathcal{D} \rightarrow \mathbb{C}$ is continuous. Then the function

$$g(z) := \int_a^b f(t, z) dt$$

is analytic in $\mathcal{D}$ and

$$g'(z) = \int_a^b \frac{\partial f(t, z)}{\partial z} dt. \quad (3.46)$$

Proof. For any fixed t , write $f(t, z) = p(t, x, y) + i q(t, x, y)$ and $g(z) = u(x, y) + i v(x, y)$. Then we have $u = \int p dt$ and $v = \int q dt$. Then the Cauchy–Riemann theorem 2.50 applied to f yields

$$\begin{aligned} \frac{\partial u}{\partial x} &= \int \frac{\partial p}{\partial x} = \int \frac{\partial q}{\partial y} = \frac{\partial v}{\partial y}, \\ \frac{\partial u}{\partial y} &= \int \frac{\partial p}{\partial y} = - \int \frac{\partial q}{\partial x} = - \frac{\partial v}{\partial x}. \end{aligned}$$

Since $\frac{\partial f}{\partial z} : [a, b] \times \mathcal{D} \rightarrow \mathbb{C}$ is continuous, $g(z)$ is totally differentiable and thus analytic by the above equations and the Cauchy–Riemann theorem. By Theorem 2.66, we have $\frac{\partial f}{\partial \bar{z}} = 0$. Then (3.46) follows from Theorem 3.81. $\square$

Theorem 3.83 (Generalized Cauchy integral formula). Every analytic function is arbitrarily often complex differentiable. Each derivative is again analytic. Furthermore,

$$f^{(n)}(z) = \frac{n!}{2\pi i} \oint_{\alpha} \frac{f(\zeta)}{(\zeta - z)^{n+1}} d\zeta, \quad (3.47)$$

where the assumptions and notation are the same as those in Theorem 3.73.

Proof. Define

$$G(t, z) := \frac{f(\zeta(t))\zeta'(t)}{\zeta(t) - z}.$$

From Theorem 3.73 we have

$$f(z) = \frac{1}{2\pi i} \oint_{\alpha} G(t, z) dt.$$

The rest of the proof follows from Lemma 3.82 and an induction on n . $\square$

Corollary 3.84 (Cauchy inequality). If $f : \mathcal{D} \rightarrow \mathbb{C}$ is analytic and $\mathcal{U}_r(c) \subset \mathcal{D}$ for some $r > 0$, then we have

$$\forall n \in \mathbb{N}^+, \quad |f^{(n)}(c)| \leq \frac{n! M(r)}{r^n}, \quad (3.48)$$

where

$$M(r) = \max_{|\zeta - c| = r} |f(\zeta)|. \quad (3.49)$$

Proof. For the circle defined in (3.40), Theorem 3.83 yields

$$\begin{aligned} |f^{(n)}(c)| &= \left| \frac{n!}{2\pi i} \oint_{\alpha} \frac{f(\zeta)}{(\zeta - c)^{n+1}} d\zeta \right| \\ &\leq \frac{n!}{2\pi} \frac{M(r)}{r^{n+1}} 2\pi r = \frac{n! M(r)}{r^n}. \end{aligned} \quad \square$$

Example 3.85. For a Jordan curve α around $\mathbf{i}$, we have

$$\oint_{\alpha} \frac{\cos z}{(z - \mathbf{i})^3} dz = \frac{2\pi \mathbf{i}}{2!} (\cos z)''|_{z=\mathbf{i}} = -\pi \mathbf{i} \cos \mathbf{i} = -\pi \frac{e^{-1} + e}{2} \mathbf{i},$$

where the last step follows from Corollary 1.94.

Exercise 3.86. Compute $I = \oint_{|z|=2\pi} \frac{z^3}{(z-\pi)^3} dz$.

Exercise 3.87. Suppose $f : \mathbb{D} \rightarrow \mathbb{C}$ is analytic. Show that the diameter $d = \sup_{z, w \in \mathbb{D}} |f(z) - f(w)|$ of the image of f satisfies

$$2|f'(0)| \leq d.$$

Hint: Show that $2f'(0) = \frac{1}{2\pi i} \int_{|\zeta|=r} \frac{f(\zeta) - f(-\zeta)}{\zeta^2} d\zeta$ for any $r \in (0, 1)$.

Exercise 3.88. Let f be a bounded analytic function on the unit disk $\mathbb{D}$. Prove that

$$\sup_{z \in \mathbb{D}} (1 - |z|) |f'(z)| < +\infty.$$

Hint: For $z \in \mathbb{D}$, use the Cauchy inequality at the circle centered at z with radius $1 - |z|$.

Exercise 3.89. Let f be an analytic function on the strip $\mathcal{D} := \{z \in \mathbb{C} : |\operatorname{Im} z| < 1\}$. For a fixed $\eta \in \mathbb{R}$, there exists $A > 0$ such that

$$\forall z \in \mathcal{D}, \quad |f(z)| \leq A(1 + |z|)^{\eta}.$$

Show that for each $n \in \mathbb{N}$, there exists $A_n > 0$ such that

$$\forall x \in \mathbb{R}, \quad |f^{(n)}(x)| \leq A_n(1 + |x|)^{\eta}.$$

Hint: Use the Cauchy inequality.

Exercise 3.90. Let f be an analytic function on the closed unit disk $\overline{\mathbb{D}}$.

(a) Prove that

$$\frac{1}{2\pi i} \oint_{|\zeta|=1} \frac{\overline{f(\zeta)}}{\zeta - z} d\zeta = \begin{cases} \overline{f(0)} & \text{if } |z| < 1, \\ \overline{f(0) - f(\frac{1}{\bar{z}})} & \text{if } |z| > 1. \end{cases}$$

(b) If $|z| < 1$, compute

$$\frac{1}{2\pi i} \oint_{|\zeta|=1} \frac{f(\zeta)}{\zeta - z} d\zeta.$$

(c) If P is a polynomial, show that

$$\oint_{|\zeta|=1} \overline{P(\zeta)} d\zeta = 2\pi i \overline{P'(0)}.$$

Hint: You can verify that

- $\int_{\alpha} \overline{f(\zeta)} d\zeta = \overline{\int_{\alpha} f(\zeta) d\bar{\zeta}}$, where α is any piecewise smooth curve.
- For $|\zeta| = 1$, it holds that $\bar{\zeta} = \frac{1}{\zeta}$, $d\bar{\zeta} = -\frac{d\zeta}{\zeta^2}$ and $d|\zeta| = \frac{d\zeta}{i\zeta}$.

3.4.2 Morera's theorem

Theorem 3.91 (Morera 1886). Let $\mathcal{D} \subseteq \mathbb{C}$ be open and $f : \mathcal{D} \rightarrow \mathbb{C}$ be a continuous function. If

$$\oint_{\langle z_1, z_2, z_3 \rangle} f(\zeta) d\zeta = 0 \quad (3.50)$$

for every triangular path $\langle z_1, z_2, z_3 \rangle$ whose triangle is entirely contained in $\mathcal{D}$, then f is analytic.

Proof. For any given point $z_0 \in \mathcal{D}$, there is an open neighborhood $\mathcal{U}_\epsilon(z_0) \subset \Delta(z_1, z_2, z_3) \subset \mathcal{D}$. It suffices to show that f is analytic in $\mathcal{U}_\epsilon(z_0)$. By (3.50), Lemma 3.56, and the fact that every polygon has a triangulation, the integral of f along any Jordan curve in $\mathcal{U}_\epsilon(z_0)$ vanishes. Theorem 3.40 then implies that f has a primitive F and the proof is completed by applying Theorem 3.83 to F . $\square$

3.4.3 Liouville's theorem

Definition 3.92. An *entire function* is an analytic function $f : \mathbb{C} \rightarrow \mathbb{C}$.

Theorem 3.93 (Liouville 1847). Every bounded entire function is a constant.

Proof. We show $f'(z) = 0$ for all $z \in \mathbb{C}$. Since f is bounded, there exists $C > 0$ such that $|f(z)| \leq C$ for all $z \in \mathbb{C}$. Then Corollary 3.84 implies that

$$\forall z \in \mathbb{C}, \forall r > 0, \quad |f'(z)| \leq \frac{C}{r}.$$

Passing to the limit $r \rightarrow +\infty$ completes the proof. $\square$

Example 3.94. $f(z) = \cos(z)$ is not bounded since

$$\lim_{x \rightarrow +\infty} \cos(ix) = \lim_{x \rightarrow +\infty} \frac{e^x + e^{-x}}{2} = +\infty.$$

Hence Theorem 3.91 does not apply to $\cos(z)$.

Exercise 3.95 (The generalized Liouville). Let f be an entire function. For a given integer $m \geq 1$, there exists $C > 0$ such that

$$\forall z \in \mathbb{C}, \quad |f(z)| \leq C(1 + |z|)^m.$$

Show that f is a polynomial of degree at most m .

Hint: Use the Cauchy inequality to show $f^{(m+1)} \equiv 0$.

Exercise 3.96. We say $f : \mathbb{C} \rightarrow \mathbb{C}$ is *doubly periodic function* if there exist $\tau \in \mathbb{C} \setminus \mathbb{R}$ such that

$$\forall z \in \mathbb{C}, \quad f(z+1) = f(z+\tau) = f(z).$$

Show that every entire doubly periodic function is a constant.

Exercise 3.97. Let f be a nonconstant entire function. Show that $f(\mathbb{C})$ is dense in $\mathbb{C}$, that is, for any $\zeta \in \mathbb{C}$, there exists a sequence $\{z_n\}$ such that $f(z_n) \rightarrow \zeta$.

Hint: Suppose there exist $\zeta \in \mathbb{C}$ and $r > 0$ such that $\mathcal{U}_r(\zeta) \cap f(\mathbb{C}) = \emptyset$. Then apply Liouville's theorem to $\frac{1}{f(z)-\zeta}$.

Exercise 3.98 (Alternative proof of the fundamental theorem of algebra). Prove Theorem 3.47 by applying Liouville's theorem.

3.4.4 The stability of complex calculus on power series

Lemma 3.99. Suppose a sequence $(f_n : \mathcal{D} \rightarrow \mathbb{C})_{n \in \mathbb{N}}$ of continuous functions with $\mathcal{D}$ open converges locally uniformly to f . Then we have

$$\lim_{n \rightarrow \infty} \int_{\gamma} f_n(z) dz = \int_{\gamma} f(z) dz, \quad (3.51)$$

where $\gamma : [0, 1] \rightarrow \mathcal{D}$ is any piecewise smooth curve.

Exercise 3.100. Prove Lemma 3.99.

Theorem 3.101 (Weierstrass 1841). Let $\mathcal{D} \subset \mathbb{C}$ be open and $(f_n : \mathcal{D} \rightarrow \mathbb{C})$ be a sequence of analytic functions. If (f_n) converges locally uniformly to f in the sense of Definition D.168, then f is analytic and the sequence of derivatives (f'_n) converges locally uniformly to f' .

Proof. Let $\gamma := \langle z_1, z_2, z_3 \rangle$ be an arbitrary triangular path whose triangle is entirely contained in $\mathcal{D}$. We have

$$\oint_{\gamma} f(\zeta) d\zeta = \oint_{\gamma} \lim_{n \rightarrow \infty} f_n(\zeta) d\zeta = \lim_{n \rightarrow \infty} \oint_{\gamma} f_n(\zeta) d\zeta = 0,$$

where the first equality follows from the local uniform convergence of (f_n) to f , the second from Lemma 3.99, and the last from Cauchy's Theorem 3.45. Then f being analytic follows from Morera's theorem 3.91.

(f'_n) converges locally uniformly to f' because

$$\forall c \in \mathcal{D}, \exists r > 0 \text{ s.t. } \forall z \in \mathcal{U}_r(c) \subset \mathcal{D},$$

$$\begin{aligned} \lim_{n \rightarrow \infty} f'_n(z) &= \lim_{n \rightarrow \infty} \left(\frac{1}{2\pi i} \oint_{\alpha} \frac{f_n(\zeta)}{\zeta - z} d\zeta \right)' \\ &= \lim_{n \rightarrow \infty} \frac{1}{2\pi i} \oint_{\alpha} \frac{f_n(\zeta)}{(\zeta - z)^2} d\zeta = \frac{1}{2\pi i} \oint_{\alpha} \lim_{n \rightarrow \infty} \frac{f_n(\zeta)}{(\zeta - z)^2} d\zeta \\ &= \frac{1}{2\pi i} \oint_{\alpha} \frac{f(\zeta)}{(\zeta - z)^2} d\zeta = f'(z), \end{aligned}$$

where α is the circle in (3.40), the first step follows from the Cauchy integral formula (Theorem 3.73), the second from the generalized Cauchy integral formula (Theorem 3.83), the third from Lemma 3.99, the fourth from the locally uniform convergence of (f_n) to f , and the last from the Cauchy integral formula. $\square$

Theorem 3.102 (Weierstrass 1841). If a series $\sum_{n=0}^{\infty} f_n$ of analytic functions $\mathcal{D} \mapsto \mathbb{C}$ with $\mathcal{D} \subseteq \mathbb{C}$ open converges normally to f in the sense of Definition 1.74, then f is analytic and the series $\sum_{n=0}^{\infty} f'_n$ converges normally to f' .

Proof. By Theorem 1.75, the normal convergence of $\sum_{n=0}^{\infty} f_n$ implies the local uniform convergence of the sequence $(\sum_{i=0}^n f_i)_{n=0}^{\infty}$ to f . By Theorem 3.101, f is analytic. For any given $c \in \mathcal{D}$ we choose $\epsilon > 0$ small enough such that the closed disk $\overline{\mathcal{U}_{2\epsilon}(c)}$ is entirely contained in $\mathcal{D}$. Then the generalized Cauchy integral formula yields

$$\forall z \in \mathcal{U}_{\epsilon}(c), \quad |f'_n(z)| = \left| \frac{1}{2\pi i} \oint_{|\zeta-c|=2\epsilon} \frac{f_n(\zeta)}{(\zeta - z)^2} d\zeta \right| \leq \frac{2M_n}{\epsilon},$$

where $\frac{1}{|\zeta - z|^2} \leq \frac{1}{\epsilon^2}$ holds for all $z \in \mathcal{U}_{\epsilon}(c)$ and the inequality follows from Lemma 3.7. The rest follows from Definition 1.74. $\square$

Corollary 3.103. The derivative of the function $f : \mathbb{C} \rightarrow \mathbb{C}$

$$f(z) = \sum_{n=0}^{\infty} a_n(z - c)^n \quad (3.52)$$

can be obtained termwise in its convergence disk $\mathcal{U}_r(c)$:

$$\forall z \in \mathcal{U}_r(c), \quad f'(z) = \sum_{n=1}^{\infty} n a_n(z - c)^{n-1}. \quad (3.53)$$

Proof. This follows from Theorems 1.77 and 3.102. $\square$

Exercise 3.104. With the assumptions of Theorem 3.101, show that for each $k \in \mathbb{N}$, $(f_n^{(k)})$ converges locally uniformly to $f^{(k)}$, and $f^{(k)}$ can be obtained termwise in the convergence disk $\mathcal{U}_r(c)$:

$$\forall z \in \mathcal{U}_r(c), \quad f^{(k)}(z) = \sum_{n=k}^{\infty} n(n-1) \cdots (n-k+1) a_n(z - c)^{n-k}.$$

Exercise 3.105. Show that if f is analytic in $\mathbb{D}$ and continuous on $\overline{\mathbb{D}}$, then $\oint_{|\zeta|=1} f(\zeta) d\zeta = 0$.

Hint: Consider a sequence of functions $(f_r)_{0 < r < 1}$, where

$$f_r : \overline{\mathcal{U}_{\frac{1}{r}}(0)} \rightarrow \mathbb{C}, \quad f_r(z) = f(rz).$$

Definition 3.106. The p -series or the *hyperharmonic series* is defined for any $p > 0$ as

$$S_p := \sum_{n=1}^{\infty} \frac{1}{n^p}. \quad (3.54)$$

It is called an *overharmonic series* if $p > 1$.

Theorem 3.107. The p -series diverges for $p \leq 1$ and converges for $p > 1$.

Proof. Use the integral test in Theorem B.50. $\square$

Definition 3.108. *Riemann zeta function* is defined as

$$\zeta(s) := \sum_{n=1}^{\infty} \frac{1}{n^s}, \quad (3.55)$$

where $s \in \mathbb{C}$ and $\operatorname{Re} s > 1$.

Corollary 3.109. The series in (3.55) converges normally and the Riemann zeta function is analytic in the half-plane

$$\mathcal{D} = \{s \in \mathbb{C} : \operatorname{Re} s > 1\}. \quad (3.56)$$

Proof. By Corollary 2.26, we have

$$|n^s| = |\exp(s \operatorname{Log} n)| = |n^{\operatorname{Re} s}|.$$

Put $M_n := \left(\frac{1}{n}\right)^{\operatorname{Re} s}$ and the proof is completed by Definition 1.74, Theorem 3.102, and Theorems 3.107. $\square$

Exercise 3.110. *Bessel function of order m* is defined as

$$\mathcal{J}_m(z) := \sum_{n=0}^{\infty} \frac{(-1)^n}{n!(m+n)!} \left(\frac{z}{2}\right)^{2n+m},$$

where $m \in \mathbb{N}$. By computing the convergence radius of the above power series, show that $\mathcal{J}_m$ is an entire function.

Exercise 3.111. The *hypergeometric series* is defined as

$$F(a, b, c; z) := \sum_{n=0}^{\infty} \frac{(a)_n (b)_n}{(c)_n} \frac{z^n}{n!},$$

where $a, b, c \in \mathbb{C}$ are fixed, $-c \notin \mathbb{N}$, and $(z)_n$ is the *Pochhammer symbol* defined as

$$\forall z \in \mathbb{C}, \quad (z)_n := \begin{cases} 1 & \text{if } n = 0, \\ z(z+1) \cdots (z+n-1) & \text{if } n \in \mathbb{N}^+. \end{cases}$$

Show that $F(z)$ converges for $|z| < 1$ and satisfies the differential equation

$$z(1-z)F''(z) + (c - (a+b+1)z)F'(z) - abF(z) = 0.$$

3.5 The ultimate characterization of analytic functions

Theorem 3.112 (Power series expansion, Cauchy 1831). Any analytic function $f : \mathcal{D} \rightarrow \mathbb{C}$ with $\mathcal{D} \subseteq \mathbb{C}$ open satisfies

$$\forall c \in \mathcal{D}, \forall z \in \mathcal{U}_R(c) \subseteq \mathcal{D}, \quad f(z) = \sum_{n=0}^{\infty} a_n(z-c)^n, \quad (3.57)$$

where $\mathcal{U}_R(c)$ is the open disk with its radius as R and its center at c and the coefficients are uniquely determined as

$$a_n := \frac{f^{(n)}(c)}{n!} = \frac{1}{2\pi i} \oint_{|\zeta-c|=\rho} \frac{f(\zeta)}{(\zeta-c)^{n+1}} d\zeta, \quad (3.58)$$

with ρ being an arbitrary real positive number in $(0, R)$.

Proof. $|z-c| < |\zeta-c|$ holds for all $z \in \mathcal{U}_\rho(c)$. Then the geometric series yields

$$\frac{1}{\zeta-z} = \frac{1}{(\zeta-c)(1-\frac{z-c}{\zeta-c})} = \sum_{n=0}^{\infty} \frac{(z-c)^n}{(\zeta-c)^{n+1}}.$$

It follows that

$$\begin{aligned} f(z) &= \frac{1}{2\pi i} \oint_{|\zeta-c|=\rho} \frac{f(\zeta)}{\zeta-z} d\zeta \\ &= \frac{1}{2\pi i} \oint_{|\zeta-c|=\rho} f(\zeta) \sum_{n=0}^{\infty} \frac{(z-c)^n}{(\zeta-c)^{n+1}} d\zeta \\ &= \sum_{n=0}^{\infty} (z-c)^n \frac{1}{2\pi i} \oint_{|\zeta-c|=\rho} \frac{f(\zeta)}{(\zeta-c)^{n+1}} d\zeta \\ &= \sum_{n=0}^{\infty} a_n(z-c)^n, \end{aligned}$$

where the first step follows from the Cauchy integral formula (Theorem 3.73), the third step from Theorems 1.77, 1.75, and D.178, and the last step from (3.58) and Theorem 3.83. Finally, the uniqueness of a_n follows from the uniqueness of the above derivation. $\square$

Exercise 3.113 (Gutzmer's inequality). With the assumptions of Theorem 3.112, show that for any $r \in (0, R)$,

$$\sum_{n=0}^{\infty} |a_n|^2 r^{2n} \leq M(r)^2, \quad (3.59)$$

where $M(r)$ is defined in (3.49).

Exercise 3.114 (Alternative proof of Corollary 3.84). Prove (3.48) by using (3.59). Furthermore, show that the equality in (3.48) holds for some $n_0 \in \mathbb{N}$ if and only if $f(z) = a_{n_0}(z-c)^{n_0}$.

Example 3.115. Theorem 3.112 applied to $\text{Log}(z)$ on $\mathbb{C}_-$ in (2.5) yields a power series expansion,

$$\text{Log}(z) = \text{Log}(c) + \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{c^n n} (z-c)^n, \quad (3.60)$$

where we have used the fact $(\text{Log} z)' = \frac{1}{z}$ (why?). By Theorem 1.79, the convergence radius is $|c|$. Interestingly, this example shows that the convergence radius may be bigger than the distance of the series center to the boundary of the domain of definition.

One needs to be careful about the convergence of a Taylor series. If $c \in \mathbb{C}_-$ satisfies $\text{Re } c < 0$, then $\mathbb{C}_- \cap \mathcal{U}_r(c)$

might consist of two disjoint separable connected components, where the power series converge to different functions,

$$f(z) = \begin{cases} \text{Log}(z) & \text{if } \text{Im } c > 0; \\ \text{Log}(z) + 2\pi i & \text{if } \text{Im } c < 0. \end{cases}$$

Example 3.116. $\sum_{n=1}^{\infty} \frac{z^n}{n^2}$ converges for $|z| = 1$, $\sum_{n=1}^{\infty} z^n$ diverges for $|z| = 1$, and the series

$$p(z) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{z^n}{n} \quad (3.61)$$

diverges for $z = -1$ and converges for all other z satisfying $|z| \leq 1$. Why? The answers lie in Theorem 3.107 and the fact that $p(z) = \text{Log}(1+z)$.

Example 3.117. Consider the geometric series

$$g(z) = \frac{1}{1-z^2} = \sum_{n=0}^{\infty} z^{2n}.$$

The divergence of $\sum_{n=0}^{\infty} z^{2n}$ for $|z| \geq 1, z \in \mathbb{R}$ is easily understood: $g(z)$ has two singularities at $z = \pm 1$. However the divergence of the geometric series

$$f(z) = \frac{1}{1+z^2} = \sum_{n=0}^{\infty} (-1)^n z^{2n}.$$

for $|z| \geq 1, z \in \mathbb{R}$ is perplexing. After all, $f(z)$ is always bounded for any given $z \in \mathbb{R}$. This puzzle is solved in the context of complex analysis by observing that $f(z)$ has two singularities at $z = \pm i$. Indeed, $f(z) = g(iz)$ and the two functions are really the same complex function with different coordinates.

Exercise 3.118 (Double series theorem of Weierstrass). Let the power series

$$f_j(z) = \sum_{k=0}^{\infty} c_{jk}(z-c)^k \quad (3.62)$$

be convergent in the disk $\mathcal{U}_r(c)$ with $r > 0$. If the series $F := \sum_{j=0}^{\infty} f_j$ converges normally in $\mathcal{U}_r(c)$, show that F is analytic in $\mathcal{U}_r(c)$ and can be represented as

$$F(z) = \sum_{k=0}^{\infty} \left(\sum_{j=0}^{\infty} c_{jk} \right) (z-c)^k. \quad (3.63)$$

Example 3.119. The following $\mathcal{C}^\infty$ function due to Cauchy

$$f(x) = \begin{cases} \exp(-\frac{1}{x^2}), & x \neq 0; \\ 0, & x = 0 \end{cases} \quad (3.64)$$

satisfies $f^{(n)}(0) = 0$ for all $n \in \mathbb{N}$, but it does not have a Taylor expansion at any neighborhood at 0.

Theorem 3.120 (Characterizations of analytic functions). The following assertions are equivalent for a function $f : \mathcal{D} \rightarrow \mathbb{C}$ with $\mathcal{D} \subset \mathbb{C}$ open.

(A) f is analytic at any point $z \in \mathcal{D}$.

(B) f is totally differentiable in the sense of real analysis ($\mathbb{C} = \mathbb{R}^2$) and satisfies the Cauchy–Riemann equations (2.31).

(C) f is continuous and for each triangle path $\langle z_1, z_2, z_3 \rangle$ whose convex hull is contained in $\mathcal{D}$ we have

$$\oint_{\langle z_1, z_2, z_3 \rangle} f(\zeta) d\zeta = 0.$$

(D) f locally admits primitives, i.e., for each point $c \in \mathcal{D}$ there exists an open disk $\mathcal{U}(c) \subset \mathcal{D}$ such that $f|_{\mathcal{U}(c)}$ has a primitive.

(E) f is continuous, and for each disk $\mathcal{U}_\rho(c)$ satisfying $\overline{\mathcal{U}_\rho(c)} \subset \mathcal{D}$ we have

$$\forall z \in \mathcal{U}_\rho(c), \quad f(z) = \frac{1}{2\pi i} \oint_{|\zeta-c|=\rho} \frac{f(\zeta)}{\zeta-z} d\zeta.$$

(F) f is locally representable as a convergent power series.

(G) f is representable as a power series in each open disk fully contained in $\mathcal{D}$.

Proof. We list the main points of the proof as follows.

- $(A) \Leftrightarrow (B)$: Cauchy–Riemann theorem 2.50;

- $(A) \Rightarrow (C)$: Cauchy theorem 3.53;
- $(C) \Rightarrow (A)$: Morera’s theorem 3.91;
- $(C) \Rightarrow (D)$: Theorem 3.40 and Lemma 3.56;
- $(D) \Rightarrow (C)$: Theorem 3.40;
- $(A) \Rightarrow (E)$: Cauchy integral formula 3.73;
- $(E) \Rightarrow (F)$: Taylor theorem 3.112;
- $(F) \Rightarrow (A)$: Corollary 3.103;
- $(A) \Rightarrow (G)$: Taylor theorem 3.112;
- $(G) \Rightarrow (F)$: A universal statement implies a existential statement. $\square$

Exercise 3.121. Let the power series $f(z) = \sum_{n=0}^{\infty} a_n z^n$ be convergent in $\mathcal{U}_r(0)$. Show that if f is nonconstant and $\sum_{n=2}^{\infty} n|a_n|r^{n-1} \leq |a_1|$, then f is injective.

Exercise 3.122. Let the power series $f(z) = \sum_{n=0}^{\infty} a_n z^n$ be convergent in $\mathcal{U}_R(0)$. Define $A(r) = \max_{|z|=r} \operatorname{Re} f(z)$. Show that for each $r \in (0, R)$ and $n \in \mathbb{N}^+$,

- (a) $a_n r^n = \frac{1}{\pi} \int_0^{2\pi} [\operatorname{Re} f(re^{i\theta})] e^{-in\theta} d\theta$;
- (b) $|a_n| r^n \leq 2A(r) - 2\operatorname{Re} f(0)$.